

Prevalence and correlates of prevalent sexually transmitted infections among young women employed as domestic workers in Central Uganda: a community-based cross-sectional study

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ABSTRACT

Introduction Little is known about the burden of sexually transmitted infections (STIs) among adolescent girls and young women (AGYW) employed as domestic workers in Ugandan households.

Methods We conducted a community-based cross-sectional study from November 2023 to March 2024 to determine the prevalence and correlates of STIs among Ugandan AGYW aged 14–24 in the Kampala Metropolitan Area. Self-collected vaginal swabs were tested for *Neisseria gonorrhoeae* (NG) and *Chlamydia trachomatis* (CT) using GeneXpert. HIV and syphilis testing were performed using the Bioline HIV/syphilis Duo test. Correlates of STI prevalence were determined using modified Poisson regression.

Results Among 262 AGYW, median age was 20 years (IQR 18–23) and 87 (33%) had an STI: CT (27.5%), syphilis (5.7%), HIV (4.6%) and NG (1.5%). Overall, 14.9% had >1 STI. Of 12 AGYW with HIV, 3 (25%) were newly diagnosed and 6/9 had detectable viral loads (≥ 1000 copies/mL). In the prior 12 months, 126 (48%) had received syndromic STI treatment, but only 35 (28%) notified their partners. Additionally, 102 (39%) had used emergency contraception and 54% believed it was protective against STIs, including HIV. Factors associated with STI prevalence included non-cash compensation for domestic work (adjusted prevalence ratio (aPR) 2.26; 95% CI 1.04 to 4.92; $p=0.040$), lower educational attainment (aPR 1.73; 95% CI 1.16 to 2.57; $p=0.007$), alcohol consumption in the past 6 months (aPR 1.72; 95% CI 1.12 to 2.64; $p=0.013$) and transactional sex during the past 6 months (aPR 1.43; 95% CI 1.01 to 2.04; $p=0.045$). Conversely, self-reported sexual abuse was negatively associated with STIs (aPR 0.48; 95% CI 0.28 to 0.83; $p=0.008$).

Conclusion The high prevalence of undiagnosed STIs and unsuppressed HIV among AGYW domestic workers in Central Uganda highlights the urgent need to increase

STI/HIV prevention, testing and treatment coverage for these underserved and vulnerable young women. Targeted interventions are needed to mitigate STI/HIV acquisition in this population.

INTRODUCTION

Adolescent girls and young women (AGYW) in sub-Saharan Africa (SSA) bear a significant burden of HIV and other sexually transmitted infections (STIs), with about 80% of global HIV acquisitions occurring among AGYW in SSA.¹ Worldwide, more than 1 million curable STIs are acquired daily, with the prevalence and associated burden relatively higher in SSA.² In Uganda, AGYW account for 32% of all new HIV acquisitions, with an estimated 730 acquisitions occurring every week.³ Data from three consecutive Uganda Demographic Health Surveys show a pooled self-reported STI prevalence of 26.0% among AGYW aged 15–24 years, with 22.0%, 36.3% and 23.1% reporting an STI infection in the years 2006, 2011 and 2016, respectively.⁴ Another study compared STI prevalence among AGYW participating in HIV prevention studies and reported a prevalence of *Chlamydia trachomatis* (CT) at 10.3% and *Neisseria gonorrhoeae* (NG) at 1.7% in Eastern Africa compared with CT at 15.1% and NG at 4.6%, respectively, in South Africa.⁵ A study among Ugandan AGYW aged 15–24 years reported NG and/or CT prevalence of 6.8% in urban areas compared with 7.8% in rural areas.⁶ The elevated rates of HIV and STIs are attributable to biological

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Adolescent girls and young women (AGYW) in Uganda who are out of school and employed in household domestic work are a highly vulnerable population, often migrating from rural communities to cities with consequent social isolation and are subject to unequal power dynamics with employers. This puts them at risk of intergenerational and transactional sex. The prevalence of sexually transmitted infections (STIs) and HIV, and the associated socio-behavioural factors, have been poorly studied in this demographic, limiting the development of interventions for this group.

WHAT THIS STUDY ADDS

⇒ This cross-sectional study, which estimated STI and HIV prevalence among young females employed in household domestic work, found that one-third of AGYW domestic workers were diagnosed with at least one STI. The findings reveal a high burden of STIs and HIV, with many experiencing co-infections. Contributing factors such as poor working conditions, limited educational attainment and prevalent sexual behaviours associated with STIs, including transactional sex, sexual abuse, alcohol use and inconsistent condom use, exacerbate their vulnerability.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ This research provides insights into STI and HIV risks for vulnerable AGYW employed in domestic work. It highlights the need for tailored HIV and STI interventions that focus on increased awareness, economic and educational empowerment, improved working conditions, socio-behavioural changes, legal redress, as well as expanded access to STI/HIV testing and treatment services for this hard-to-reach population.

vulnerabilities, including the immaturity of cervical epithelial tissue, large cervicovaginal mucosal surface area and prolonged exposure to HIV at mucosal sites.⁷

The management of STIs in SSA mainly depends on a syndromic approach due to limited access to aetiological testing. This method fails to detect asymptomatic infections, especially in women, leading to many undiagnosed and untreated patients, which in turn fuel ongoing transmission and long-term clinical sequelae.⁸ A meta-analysis of STI studies from low- and middle-income countries found that the overall proportions of women with asymptomatic CT, NG and *Trichomonas vaginalis* (TV) infections were 60.7%, 53.3% and 56.9%, respectively. These rates were highest among African women with CT, NG and TV at 68.6%, 67.2% and 64.9%, respectively.⁹

Behavioural vulnerabilities for AGYW include age-disparate relationships that often lead to engagement in high-risk sexual behaviours, such as inconsistent condom usage and transactional sex, compounded by a limited capacity to recognise dangerous situations and adopt protective measures.¹⁰ Furthermore, AGYW face gender-based discrimination and stigma intensified by prevailing societal norms,¹¹ which further increase their vulnerability to HIV and other STIs. Simultaneously, access to healthcare remains limited.¹⁰

Girls in SSA frequently leave school due to cultural traditions like early marriage, gender-based expectations

including household responsibilities from an early age, the belief that they do not need as much education as boys, financial hardships like unaffordable school fees and poor menstrual hygiene management.¹² Without access to formal skills training, out-of-school AGYW tend to gravitate toward low-wage jobs, such as domestic work. Research indicates that AGYW employed in these low-paying jobs are at an increased risk of acquiring STIs than those who are unemployed, likely due to engagement in risky sexual behaviours and limited ability to negotiate safer sex practices with unequal gender and power dynamics. These vulnerabilities, often compounded by structural factors such as poverty, limited livelihood opportunities and social marginalisation, may increase the reliance on transactional or age-disparate sexual relationships.¹³

The Determined, Resilient, Empowered, AIDS-Free, Mentored and Safe (DREAMS) partnership was specifically designed to address these interconnected drivers of HIV and STI risk among AGYW, particularly those who were out-of-school.¹⁴ By integrating sexual and reproductive health (SRH) services with life-skills training, gender empowerment, mentoring and economic strengthening interventions, DREAMS sought to reduce risky sexual behaviours while enhancing the agency and capacity of AGYW to negotiate safer sex. In doing so, the programme directly targeted the behavioural and structural pathways through which out-of-school status contributed to increased STI vulnerability.

However, within this broad category of out-of-school girls, important subpopulations remain understudied. Particularly, little is known about the burden of STIs among the subset of out-of-school girls employed in household domestic work. This group may face unique vulnerabilities, including economic dependence, social isolation, limited autonomy and increased risk of exploitation. Understanding STI risk among domestic worker AGYW is therefore critical to informing targeted prevention strategies and assessing whether existing interventions adequately reach and address the needs of this marginalised subgroup. This community-based cross-sectional study aimed to assess the prevalence and correlates of STIs among AGYW employed in household domestic work in two districts in Central Uganda.

METHODS**Study design and setting**

We conducted a community-based cross-sectional study from November 2023 to March 2024 in six urban and semi-urban communities within the Kampala Metropolitan Area (ie, Kampala and Wakiso districts). This is the most urbanised and densely populated region in Uganda, characterised by a mix of formal residential areas, informal settlements and peri-urban communities with high population mobility and socioeconomic disparities. SRH services, including STI management, are delivered through public health facilities, private

pharmacies and clinics and non-governmental organisations. Although STI services, integrated within broader HIV and SRH programmes at public health facilities, are considered free of charge, intermittent stock-outs, long waiting times, diagnostic challenges, concerns about confidentiality and transport costs may limit effective access. Urban outreach initiatives and youth-focused campaigns have used community engagement and media platforms to promote STI prevention and sexual health awareness. Despite the relatively high concentration of health facilities within the Kampala Metropolitan Area, access remains uneven, particularly for vulnerable populations such as domestic workers, who may face restricted mobility, limited autonomy, stigma and time constraints that hinder timely utilisation of STI services.^{15 16}

Study activities were conducted in the offices of community-based healthcare workers, locally known as Village Health Teams (VHTs), or in designated DREAMS safe spaces located in the local communities of Kasangati Town Council, Nansana division, Nabweru, Wakiso Town Council and Rubaga and Kawempe divisions in the two districts. The VHTs are non-professional, community-based health volunteers trained by the Ministry of Health to serve as the first point of contact for healthcare at the community level. They promote health, prevent disease and connect individuals to formal health services.

Study population

We screened and enrolled AGYW employed as household domestic workers, that is, those performing household tasks such as cooking, cleaning, laundry, childcare and gardening. We recruited females aged 14–24 years. Those <18 years had to qualify as emancipated minors, that is, pregnant, married, had a child or were self-sufficient or employed, according to national research guidelines.¹⁷ We included AGYW who had worked for at least 3 months in domestic work, had ever had penetrative vaginal sex for ease of vaginal swab self-sample collection and who were willing to respond to the questionnaire and provide biological samples (vaginal swabs and blood). We collaborated with community health workers and women's councillors to identify potential participants within their jurisdictional areas, using household registration lists of domestic workers and a door-to-door approach. They briefly explained the purpose of the study and referred the AGYW to the nearest study site, where the research team assessed their eligibility. Trained research nurses obtained written informed consent from eligible participants in either the local language, Luganda, or English. We maintained a de-identified enrolment log to track the proportion of persons who declined to enrol and documented reasons for non-participation.

Data collection and procedures

Two research assistants administered a semi-structured, pre-tested questionnaire, guided by the constructs of the Socio-Ecological model in either Luganda or English (online supplemental file 1). Questions were adapted

from a previous study and refined to reflect the socio-cultural context of the study population.¹⁸ Before data collection, subject matter experts and local leaders assessed the questionnaire for validity, reliability and cultural appropriateness, resulting in necessary modifications. We undertook a pilot study with five respondents at one study site to ensure that questions were clear and easily understood. Questions considered ambiguous, unclear or likely to cause biased responses were removed. The questionnaire included neutral open-ended questions and short-term recall (3–12 months) to reduce social desirability and recall bias regarding sexual behaviours. Participants were assigned anonymous identifiers to protect their identities. Data were recorded on tablet computers using Kobo Toolbox at study sites and backed up daily on a secure, access-restricted server. Automatic range and consistency checks were implemented during data entry, ensuring all questions required a response.

Sample collection and processing

We provided clear, step-by-step instructions using visual aids and verbally for vaginal swab self-collection to ensure a safe and hygienic process. Research nurses collected venous blood. HIV and syphilis testing were done at the point of care using the Bioline HIV/syphilis Duo test (Abbott, Illinois, USA), with STAT-PAK (Chembio, Medford, New York, USA) and Uni-Gold HIV-1/2 (Trinity Biotech, Wicklow, Ireland) used to confirm positive HIV test results, according to the Ugandan testing algorithm guidelines. Clearly labelled samples were transported in cooler boxes to the IDI Translational Laboratory within 6 hours of collection and tested for NG and CT using the Cepheid GeneXpert molecular test (Cepheid Europe, HBDC, Maurens-Scopont, France). A reactive Bioline syphilis treponemal antibody was confirmed using the rapid plasma reagin (RPR) (Becton Dickinson-Sparks, Maryland, USA). All participants with positive STI results received standard antimicrobial treatment for curable STIs, and those diagnosed with HIV were linked to antiretroviral therapy (ART) following Uganda guidelines.¹⁹ Those requiring counselling or legal services for sexual and gender-based violence were referred to professional organisations.

Sample size and statistical analysis

Sample size consideration

The sample size of 262 AGYW was calculated using Leslie Kish's formula for cross-sectional prevalence studies. We assumed an expected STI prevalence of 15%, based on findings from a South African study.²⁰ The calculation was based on a 95% confidence level, a 5% margin of error (precision) and a significance level of $\alpha=0.05$. We adjusted the calculated sample size for design effect to account for potential clustering effects, assuming an intracluster correlation coefficient of 0.0045, given that participants were drawn from six different centres/clusters that could have potentially varying behavioural patterns and risk. The sample size was inflated by 10% to

account for potential non-response. At sample size estimation, we assumed a pooled prevalence of 15% across all sites and used a proportionate-to-size allocation strategy to determine the number of participants to be recruited from each of the six clusters. Participants were systematically selected within each cluster using a predetermined sampling interval until the required sample size per site was reached.

Outcome and covariates definitions

The primary outcome was the prevalence of laboratory-confirmed CT, NG, syphilis and HIV, analysed as a composite variable. STI prevalence was defined as the proportion of AGYW who tested positive for any STI. Sexual behaviours that increased vulnerabilities to STI acquisition were defined as any of the following: sexual debut at age <18 years (age of consent under Ugandan law), having >1 sex partner, no condom use at last sex, sometimes/never condom use, emergency contraception use (a proxy measure of condomless sex), alcohol or drug use, transactional sex and sexual abuse. Sexual abuse was defined as non-consensual vaginal, anal or oral sex.

Statistical analysis

Participants' characteristics were described and presented using frequencies and percentages for categorical and median and IQR for continuous variables. χ^2 or Fisher's exact test was used to compare STI prevalence across participant characteristics. Modified Poisson regression was used to examine factors associated with STIs and estimate adjusted prevalence ratios (aPRs) and 95% CIs. Variables in the unadjusted model with $p > 0.2$ were included in the adjusted model. All statistical analyses were conducted using Stata V.17 (College Station, Texas, USA). All study results are reported according to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines (online supplemental file 2).

Patient and public involvement statement

Our study participants and selected members of the public were involved in the design and conduct of this study. Local leaders, AGYW and VHTs contributed to the study design by reviewing the questionnaire to ensure that the survey questions were clear, culturally appropriate and addressed relevant health concerns. Local community leaders and VHTs advised on recruitment strategies and facilitated participant engagement. Study findings were disseminated through community meetings and at the STI & HIV 2025 World Congress conference to promote awareness and inform future sexual health interventions among this vulnerable population.

RESULTS

We screened 306 participants and enrolled 262 (85.6%). Reasons for non-enrolment included being in school ($n=14$), fear of HIV testing ($n=9$), failure to complete study activities ($n=7$), fear of vaginal swab self-collection

($n=6$), lack of time for study activities ($n=4$) and being ≥ 24 years old ($n=4$). The median age was 20 (IQR 18–23), and 29% were ≤ 18 years old (online supplemental table 1). Most (61%) had attained a primary educational level (≤ 7 years of schooling), and 5% were married. Nearly all (96%) were formally employed, 84% were live-in domestic workers who were paid in cash and 73% earned a monthly salary between US\$14 and US\$27. Most (70%) AGYW reported sexual debut before age 15 years, while 27% had ≥ 2 sexual partners in the past 6 months. Only 28% used condoms during their last sexual encounter, and 9% reported consistent use. Over half (56%) engaged in transactional sex, 39% used emergency contraception and 11% used alcohol. A significant portion (21%) reported sexual abuse, for which 55% did not seek help due to fear of job loss or threats; employers were the most common perpetrators (63%). Overall, 75% of participants reported ≥ 2 high-risk behaviours, and nearly 40% had at least four. Only 16% of participants had been engaged in the DREAMS programme (table 1).

STI prevalence

In total, 87 AGYW (33%) had at least one STI, including HIV. STI prevalence was as follows: CT at 27.5% (72/262), syphilis at 5.7% (15/262), with 60% (9/15) having RPR titres $\geq 1:8$, HIV at 4.6% (12/262) and NG at 1.5% (4/262) (figure 1). Among those diagnosed with an STI, 14.9% (13/87) had more than 1 infection; specifically, 10 AGYW had 2 STIs, while 3 had 3 STIs. Of the 10 participants with dual STIs, 4 presented with syphilis and chlamydia, 2 had chlamydia and gonorrhoea, 3 had HIV and chlamydia and 1 had HIV and syphilis. Among the three individuals who tested positive for three STIs, two had HIV, syphilis and chlamydia; the remaining participant had chlamydia, syphilis and gonorrhoea. All nine participants known to be living with HIV reported being on ART. However, only one-third (33.3%, $n=3$) indicated that their last test results showed an undetectable viral load; the remaining six did not know their most recent viral load result. Notably, 25% (3/12) of those living with HIV were unaware of their diagnosis before study participation.

Factors associated with selected STIs and HIV

Receipt of non-cash remuneration (ie, food, clothing and lodging instead of salary) (aPR 2.26; 95% CI 1.04 to 4.92; $p=0.040$), lower educational attainment (≤ 7 years of schooling) (aPR 1.73; 95% CI 1.16 to 2.57; $p=0.007$), alcohol consumption in past 6 months (aPR 1.72; 95% CI 1.12 to 2.64; $p=0.013$) and engagement in transactional sex during the past 6 months (aPR 1.43; 95% CI 1.01 to 2.04; $p=0.045$) were positively associated with STI prevalence. In contrast, reported sexual abuse within the past 12 months was negatively associated with STIs (aPR 0.48; 95% CI 0.28 to 0.83; $p=0.008$).

DISCUSSION

This cross-sectional study found that one-third of AGYW domestic workers were diagnosed with at least one STI,

Table 1 Factors associated with sexually transmitted infections among adolescent girls and young women employed in domestic work

Variable	STI prevalence n/N (%)	Unadjusted PR (95% CI)	P value	Adjusted PR (95% CI)	P value
Overall	87/262 (33.2)	N/A		N/A	
Age					
≤18 years	23/77 (29.9)	Ref		–	–
>18 years	64/185 (34.6)	1.16 (0.78 to 1.72)	0.47	–	–
Education level					
Post primary	25/103 (24.3)	Ref		Ref	
Primary and below	62/159 (39.0)	1.61 (1.08 to 2.38)	0.018	1.73 (1.16 to 2.57)	0.007
Remuneration					
Cash remuneration	82/252 (32.5)	Ref		Ref	
Non-cash remuneration	5/10 (50.0)	1.54 (0.80 to 2.93)	0.19	2.26 (1.04 to 4.92)	0.040
Age at sexual debut					
≤12 years	26/79 (32.9)	0.97 (0.59 to 1.59)	0.90		
13–15 years	39/105 (37.1)	1.09 (0.69 to 1.73)	0.71	–	–
>15 years	5/28 (17.9)	0.53 (0.22 to 1.27)	0.15	–	–
Not sure	17/50 (34.0)	Ref		–	–
Condom use at last sex					
Yes	19/72 (26.4)	0.74 (0.48 to 1.13)	0.166	0.70 (0.45 to 1.06)	0.094
No	68/190 (35.8)	Ref		Ref	
Transactional sex in the past 6 months					
Yes	56/146 (38.4)	1.44 (1.00 to 2.07)	0.053	1.43 (1.01 to 2.04)	0.045
No	31/116 (26.7)	Ref		Ref	
Alcohol use in the last 6 months					
Yes	16/29 (55.2)	1.81 (1.24 to 2.65)	0.002	1.72 (1.12 to 2.64)	0.013
No	71/233 (30.5)	Ref		Ref	
Sexual abuse in the past 12 months in domestic work					
Yes	11/56 (19.6)	0.53 (0.30 to 0.93)	0.027	0.48 (0.28 to 0.83)	0.008
No	76/206 (36.9)	Ref		Ref	

Participants with a confirmatory laboratory result of STIs, that is, HIV, syphilis, chlamydia or gonorrhoea, were considered as having an STI. Ref is the reference category for a covariate in the modified Poisson model fitted on 262 complete cases. Variables for the multivariate model were selected using a 0.2 p value cut-off in the unadjusted analysis of each covariate. None of the variables in the adjusted model had missing values.

N, total study sample size; n, number of participants with at least one STI; PR, prevalence ratio; STIs, sexually transmitted infections.

and 15% of them were diagnosed with more than one STI. Moreover, our findings indicate an HIV prevalence of 4.6%, with 25% previously undiagnosed. This is two times the 2.2% prevalence reported by the Uganda AIDS Commission, among the general population of AGYW in Uganda.³ The overall prevalence of STIs and sexual behaviour that increase risks for STIs among AGYW domestic workers aligns with other key and priority populations in Uganda, including sex workers²¹ and AGYW participating in an HIV pre-exposure prophylaxis trial.²² The substantial STI burden in this study reflects the limited utilisation of STI services in urban Uganda, consistent with previous findings.^{15 16} In contrast, countries such as the USA and Brazil have comprehensive legal frameworks, youth-friendly services and universal

access to STI prevention and treatment, supported by explicit policies protecting adolescent SRH, confidential care and integrated sexual health education.²³

Although Uganda has made progress through initiatives like the DREAMS programme and the National Adolescent Health Policy and SRH strategy, these interventions often emphasise HIV management, leaving non-HIV STIs unaddressed. Access remains uneven, particularly for underserved populations like domestic workers, who face restricted mobility, stigma and limited autonomy.¹⁶ Despite Uganda being a signatory to the Joint United Nations Programme on HIV/AIDS (UNAIDS) 95–95–95 targets and aligning its national HIV strategy with global frameworks, structural and policy gaps continue to limit effective STI and HIV prevention and treatment for

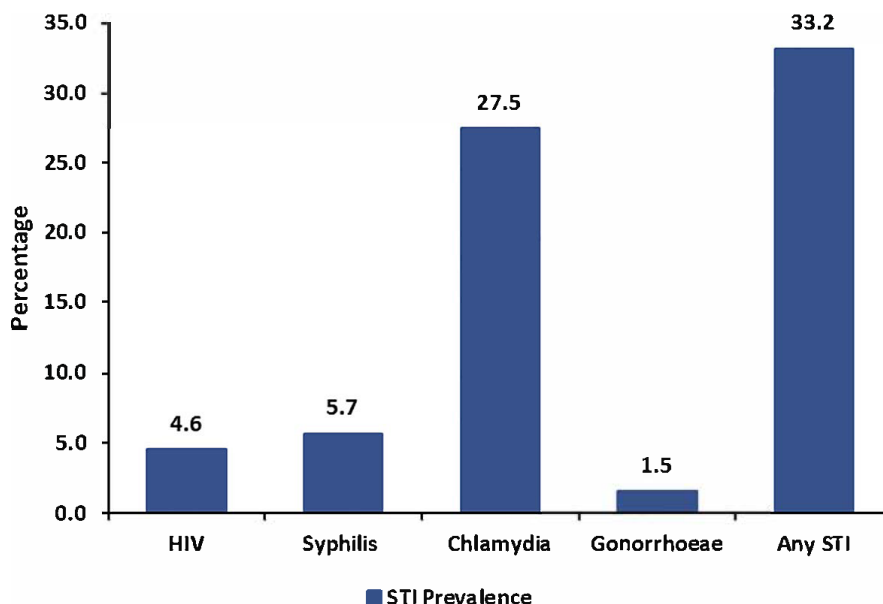


Figure 1 Prevalence of STIs among 262 adolescent girls and young women employed in domestic work in Central Uganda, 2023–2024. Chlamydia was the most common STI (27.5%), followed by syphilis (5.7%) and HIV (4.6%). Gonorrhoea had the lowest prevalence (1.5%). Overall, one in three participants (33.2%) tested positive for at least one STI. STI, sexually transmitted infection.

AGYW, findings similar to many SSA countries.²⁴ This underscores the need for expansion of youth-friendly, confidential and accessible STI services, integration of non-HIV STIs screening into existing programmes and implementation of policies that specifically address the unique vulnerabilities of out-of-school AGYW, like those employed in domestic work.

The vulnerabilities faced by these AGYW may be exacerbated by power imbalances between employers and employees, which contribute to economic exploitation, social isolation and limited access to healthcare or legal redress.¹³ Previous studies show that AGYW frequently change households, seeking better working conditions or engage in high-risk sexual behaviours, increasing their vulnerability,¹³ but also potentially contributing to STI transmission. This notable STI prevalence among AGYW underscores the need to prioritise STI control in public health initiatives in this population as a fundamental step to achieving epidemic HIV and STI control in this important but underserved demographic.

We observed prevalent sexual behaviours compounded by known factors such as low educational attainment²⁵ and unstable employment conditions, influenced by existing power dynamics in domestic work, similar to previous research.¹³ Education is known to be associated with better SRH outcomes, with female adolescents who are out of school being at higher risk of acquiring HIV and other STIs.²⁶ Policies that support girls to remain in school for longer could contribute to reducing their risk of STI acquisition.

Young women who received non-cash compensation, such as food, clothing and accommodation, and those involved in transactional sex had a higher STI prevalence than those paid in cash, consistent with previous

findings.²⁷ Mirroring other studies, we also noted an association between alcohol consumption and STI diagnosis.²⁸ This underscores the importance of improving social support, promoting economic empowerment and improving employment conditions as effective strategies for preventing STIs and HIV in this group of young women.

Previous research shows that the DREAMS programme resulted in a reduction of HIV incidence and STI symptoms among AGYW across SSA.²⁹ Only 16% of our participants had previously engaged in the DREAMS programme, indicating limited reach. Innovative strategies are needed to expand access to its core components, including income-generating activities, substance use prevention (including alcohol control), peer support, safe spaces and economic empowerment. These interventions have shown benefit in prior studies but may remain inaccessible to vulnerable populations, like AGYW employed in household domestic work. Expanding coverage of such interventions should be prioritised within broader HIV prevention efforts for AGYW in SSA.²⁹

Our research revealed a negative association between reported sexual abuse in the past year and STI prevalence, which differs from findings of earlier studies.³⁰ The underlying cause of this unusual observation remains uncertain. We hypothesise that it could be linked to unmeasured residual confounding, for example, condom use by perpetrators to avoid unwanted pregnancy, survivors receiving post-rape care, avoidance of sexual activity stemming from prior sexual trauma, under-reporting of sexual abuse due to stigma and social desirability and measurement biases.

The strengths of our study include the proportionate-to-size and systematic sampling strategies, which

contributed to a robust study design and methodology, yielding valid results that may be generalisable to other AGYW domestic workers in this setting. Additionally, we conducted reference laboratory STI testing, which enabled us to correlate self-reported sexual behaviours with aetiological STI diagnosis. The limitations of this cross-sectional study include the inability to determine temporal relationships, thus limiting our analysis of behavioural correlates associated with STI diagnoses. We did not assess STI symptoms and how they correlate to STI positivity, nor did we test for some STIs which are likely to be common in this population, for example, *Trichomonas vaginalis*, *Human papilloma virus* and *Mycoplasma genitalium*. Our study was susceptible to recall bias and relied on self-reported data for sexual behaviour. This could have introduced response and social desirability biases, although we attempted to minimise this by having same-sex, same-age-group interviewers.

CONCLUSIONS

Our research with Ugandan AGYW employed in domestic work revealed a high prevalence of STIs, often with multiple co-infections. These results highlight the urgent need to strengthen initiatives aimed at keeping girls in school, improving their working conditions and providing expanded access to STI/HIV testing and treatment. Prioritising this vulnerable group is essential to curtail further STI transmission and acquisition.

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Competing interests None declared.

Patient and public involvement Patients and/or the public were involved in the design, or conduct, or reporting, or dissemination plans of this research. Refer to the Methods section for further details.

Patient consent for publication Not applicable.

Ethics approval The study received ethical approval from Makerere University School of Public Health Research Ethics Committee, with a reference number MaKSPH-REC 262, and administrative clearance from the Uganda National Council for Science and Technology, HS3152ES. Written informed consent was obtained from all participants in Luganda or English, with impartial witnesses used for those deemed illiterate. Participants aged 15–17 years (below the legal age of consent) provided written assent before study procedures. Those who were considered emancipated minors (eg, married, pregnant, parenting, or financially independent) provided written informed consent as recommended by existing national research guidelines.¹⁷ Participants' information was strictly kept confidential, with unique study identifiers used. Data were stored in cloud-based, password-protected electronic databases and secure cabinets accessible to authorized/authorised study staff. All participants received a reimbursement of USD US\$5 in accordance with the ethics committee guidance. All participants diagnosed with curable STIs received treatment provided by the study, while those diagnosed with HIV were linked to antiretroviral therapy (ART) according to national treatment guidelines. Study findings are presented in aggregate form that prevents the identification of individual participants.

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Data availability statement Data are available upon reasonable request. The data supporting the findings of this study are available from the corresponding author upon reasonable request. The dataset has been de-identified to protect participant confidentiality and is not publicly available due to ethical and privacy restrictions imposed by the Research Ethics Committee and the Uganda National Council for Science and Technology. Researchers who meet the criteria for access to confidential data may request access by contacting the corresponding author, Annet A Onzia, at aonzia@idi.co.ug.

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